IEC 61131-3

Industrial Control Systems · SCADA Hub

COURSE DESCRIPTION

IEC 61131-3 is the international standard defining the five PLC programming languages used across every major automation vendor. This course covers Structured Text, Ladder Diagram, Function Block Diagram, Sequential Function Chart, and Instruction List; the program organization unit hierarchy; and data type management across vendor implementations. Exercises use CODESYS, a vendor-neutral IDE that implements the full specification.

PREREQUISITES

No prior programming experience required. Familiarity with basic electrical or control concepts is helpful.

COURSE TOPICS

1. **Introduction:** The pre-standard PLC landscape, what the five-language specification unified, and what it deliberately left vendor-specific.
2. **Data Types:** BOOL, INT, DINT, REAL, TIME, STRING, and user-defined structures; type mismatch consequences in live systems.
3. **Structured Text:** Pascal-style syntax — IF/THEN, FOR, WHILE, CASE — and appropriate use cases among the five languages.
4. **Ladder Diagram:** Contact/coil notation, rung structure, and reading existing automation code in the industry's most common language.
5. **Function Block Diagram:** Data-flow graphical programming, block connectivity, and use cases in PID and signal conditioning.
6. **Sequential Function Chart:** Steps, transitions, and action qualifiers for modeling processes with discrete ordered states.
7. **Program Organization Units:** Functions, Function Blocks, Programs, and Configurations; reusable library design across vendors.
8. **RTAC Integration:** IEC 61131-3 in ACSELERATOR Architect and translation from PLC programming to substation automation logic.
9. **Troubleshooting:** Implicit type conversion, scan cycle overruns, latching coil failures, and silent runtime faults.
10. **Practice Lab:** PID controller in Structured Text, motor starter in Ladder, batch sequence in SFC, analog scaling Function Block.

LEARNING OUTCOMES

- Write and debug programs in Structured Text, Ladder Diagram, and Sequential Function Chart.
- Design reusable Function Blocks with typed interfaces for use across multiple vendor platforms.
- Interpret and modify existing Ladder Diagram code in a production automation system.
- Configure a PID controller using the standard IEC 61131-3 function block interface.
- Diagnose scan cycle overruns, type conversion errors, and silent runtime faults in live PLC programs.

REQUIRED SOFTWARE

- **CODESYS** (free, Windows/Linux) — IEC 61131-3 compliant IDE with soft PLC runtime for all lab exercises.
- **PLCopen Function Block Library** (free PDF) — normative reference for standard function block interfaces across vendors.